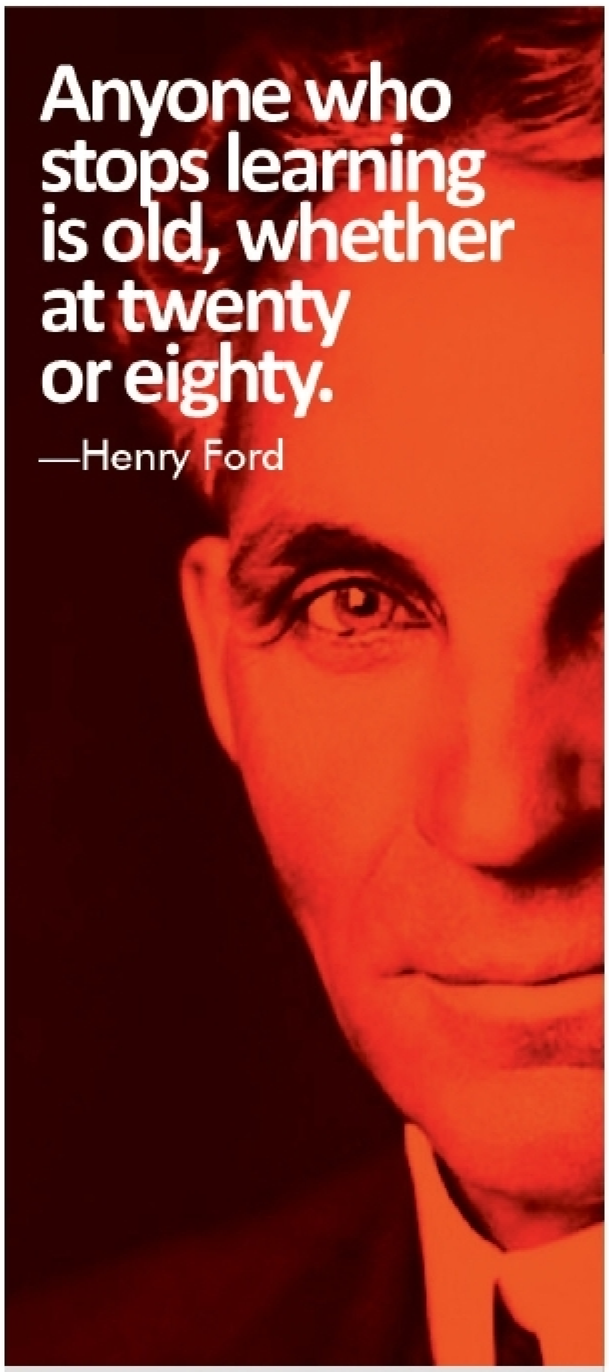

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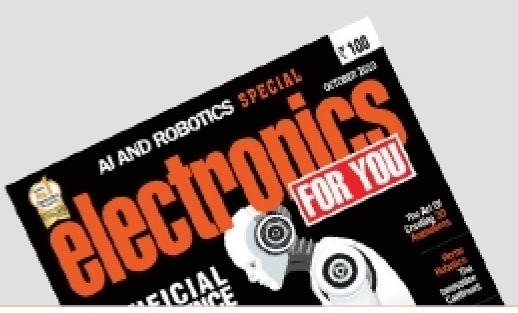

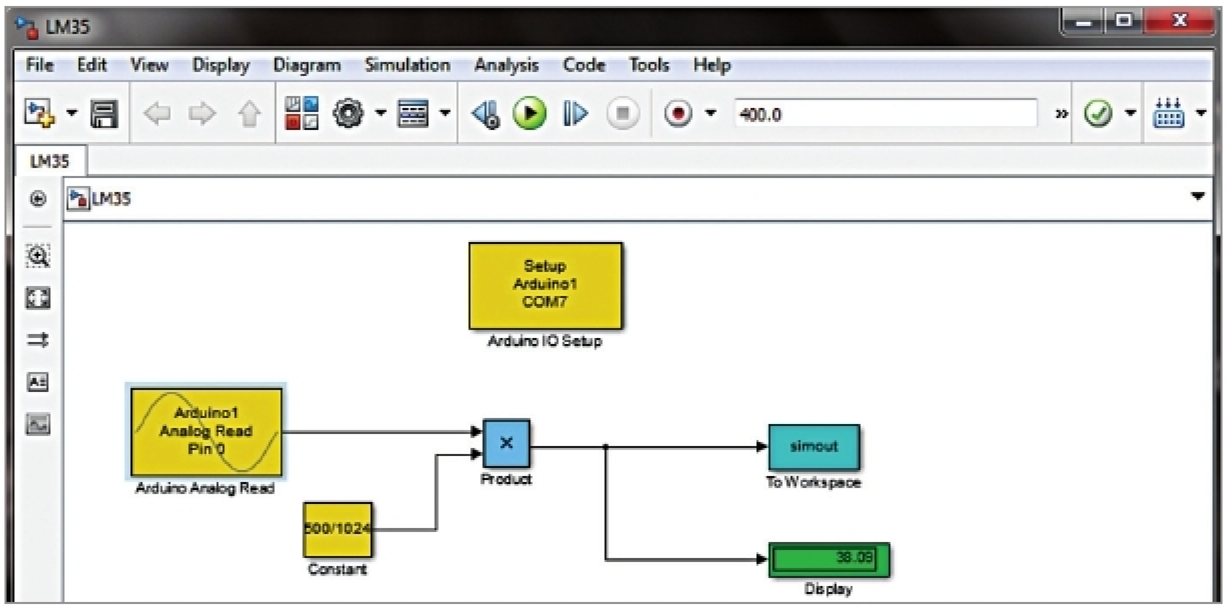

Fig. 3: Screenshot of the Simulink model

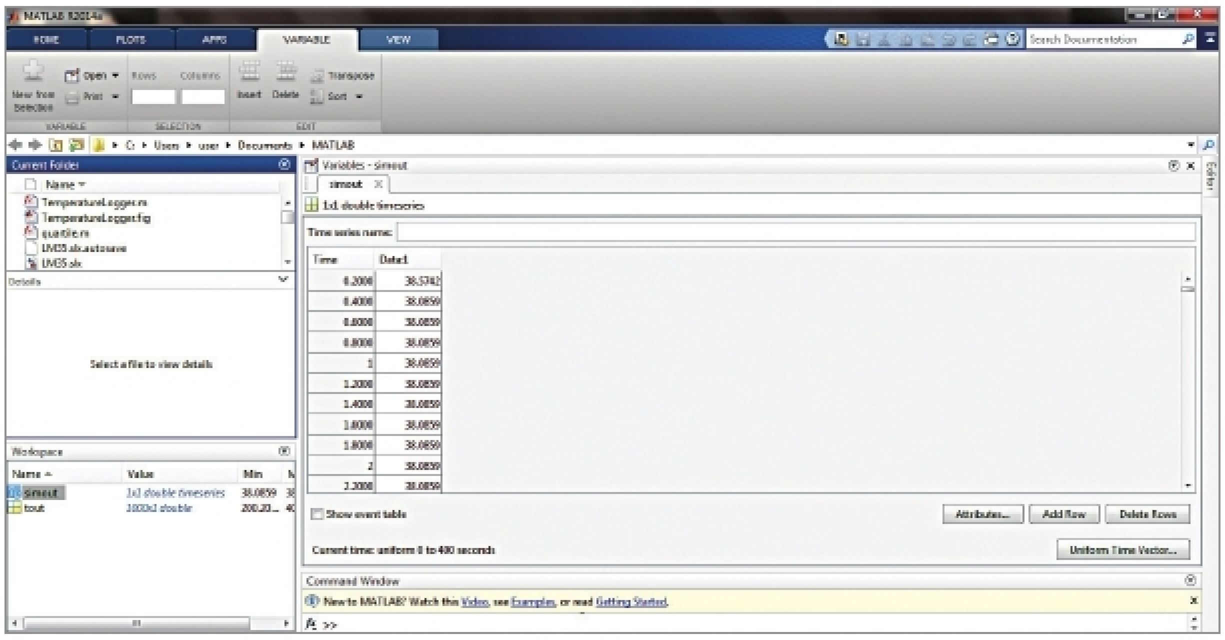

Fig. 4: Screenshot of the temperature data stored in MATLAB workspace

Uno and select COM Port number noted earlier

- Upload adios code to the Arduino Uno board by pressing the upload button in IDE
- Copy the entire contents of the extracted ArduinoIO folder to a folder in My Documents (For Windows PC). This step may be different if you are using the latest version of MATLAB
- Open Simulink LM35.slx source code file from MATLAB. The screenshot of the designed Simulink model is shown in Fig. 3
- Double click on the block 'Arduino IO Setup' and change the serial (COM) port number (set as 'COM7' here) with the corresponding port number in your PC where the Arduino has been installed
- If you want to connect LM35 to some other analogue pin of Arduino, double click on 'Arduino Analogue Read' block and set the corresponding pin
- Set 'Simulation Stop Time' as per your requirement
- Click on 'Run' button in Simulink
- The corresponding value of temperature can be seen on the 'Display' block in the model window. Additionally, the temperature data is also logged in MATLAB workspace variable 'simout'. When the simulation terminates, the temperature data can be read by double-clicking on 'simout' in the workspace window of MATLAB. The screenshot of the temperature (in degC) data saved in MATLAB workspace is shown in Fig. 4 EFY

Shibendu Mahata is MTech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. He has made several contributions in the domain of signal processing in reputed international journals and conferences

EFY Note

The source code of this project is available for free download at *source.efymag.com*